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cellular_raza: Cellular Agent-based Modeling from a Clean Slate

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Summary

`cellular_raza` is a cellular agent-based modeling framework which allows researchers to construct models from a clean slate. In contrast to other agent-based modeling toolkits, `cellular_raza` was designed to be free of assumptions about the underlying cellular representation. This enables researchers to build up complex models while retaining full control over every parameter introduced. It comes with predefined building blocks for agents and their physical domain to quickly construct new simulations bottom-up. Furthermore, `cellular_raza` can be used with the `pyo3` and `maturin` packages and thus act as a numerical backend to a python package.

Statement of Need

In contrast to classical particle simulations, ABMs treat every cell individually. This implies that parameters can vary between agents and that every cell should be traceable throughout time and space. In addition, they can describe growth, proliferation, death and many other cellular processes and should also accurately model cell lineage. These models live on the mesoscopic scale where the underlying complexity of the problem can often neither be fully attributed to intracellular or extracellular processes. Their applications include modeling of self-organization and emergent phenomena but they can also be used to introduce spatial effects into existing population-based models.

State of Field

Agent-based models (ABMs) have become popular in cellular biology (Cess & Finley, 2022; Delile et al., 2017a, 2017b; Mogilner & Manhart, 2016). While these tools have proven to be effective for targeted research questions, they often lack the ability to be applied more generically.
General-purpose agent-based toolkits on the other hand are designed without specific applications in mind (Abar et al., 2017; Datseris et al., 2022; Wilensky, 1999). They are often able to define agents bottom-up and can be a good choice if they allow for the desired cellular representation. However, they lack the explicit forethought to be applied in cellular systems and may not be able to describe every cellular aspect. In order to address these issues and construct models from first principles without any assumptions regarding the underlying complexity or abstraction level, we developed `cellular_raza`.

Cellular Agent-Based Frameworks

In our previous efforts (Pleyer & Fleck, 2023), we assessed the overall state of modelling toolkits for individual-based cellular simulations. These frameworks are designed for specific usages

and often require many parameters which are unknown or difficult to determine experimentally. This poses an inherent problem for their applicability and the ability to properly interpret results. Few modelling frameworks exist that provide a significant degree of flexibility and customization in the definition of cell agents. Chaste (Cooper et al., 2020) allows reuse of individual components , such as ODE and PDE solvers, but is only partially cell-based. Biocellion (Kang et al., 2014) supports different cell shapes such as spheres and cylinders, but admits that their current approach lacks flexibility in the subcellular description. BioDynaMo (Breitwieser et al., 2021) offers some modularity in the choice of components for cellular agents, but cannot deeply customize the cellular representation.

cellular_raza

We distinguish between different simulation aspects, i.e., mechanics, interaction, or cell cycle. These aspects are directly related to the properties of the cells, domain, or other external interactions. The user selects a cellular representation, which can be built from pre-existing building blocks or fully customized bottom-up , if desired. 'cellular_raza' utilizes macros to generate code contingent on the simulation aspects . It makes extensive use of generics and provides abstract numerical solvers. 'cellular_raza' encapsulates the inherent complexity of the code generation process, yet enables users to modify the specifics of the simulation through the use of additional keyword arguments . Consequently, users are able to fully and deeply customize the representation and behaviour of the agents. Each simulation aspect is abstractly formulated as a trait in Rust's type system. The getting-started guide provides a good entry point and explains every step from building, running to visualizing.

Examples

In the following, we present four different examples how to use `cellular_raza` (see cellular-raza.com/showcase).

Cell Sorting

Cell sorting is a naturally occurring phenomenon (Graner & Glazier, 1992; Steinberg, 1963). The cellular `Interaction` is specific to their species. We consider two distinct types represented by soft spheres. They physically attract each other at close proximity if their species is identical. Cells are placed randomly inside a cube with reflective boundary conditions. In the final snapshot, we can clearly see the phase-separation between the different species.

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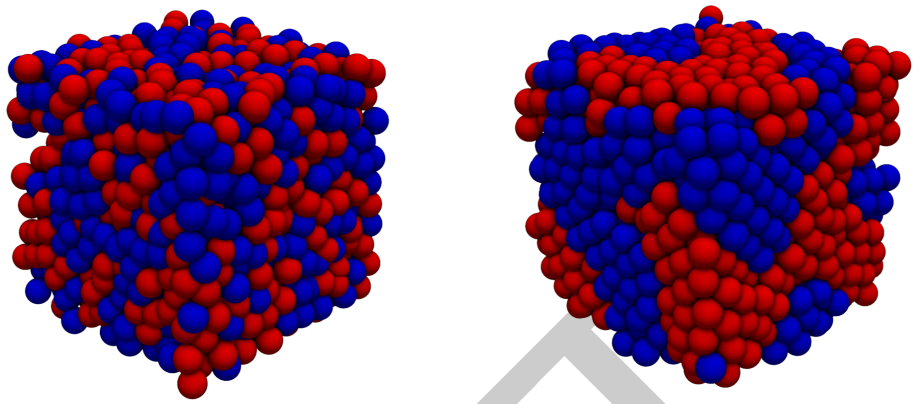

Figure 1: The initial random placement of cells reorders into a phase-separated spatial pattern.

Bacterial Rods

Bacteria come in various forms (Young, 2006; Zapun et al., 2008) such as elongated shapes (Billaudeau et al., 2017) which grow asymmetrically in the direction of elongation. Our model describes the physical mechanics of one cell as a collection of multiple vertices $\vec{v}_i$ which are connected by springs. Their relative angle α at each connecting vertex introduces a curvature force which is proportional to $2\tan(\alpha/2)$. Cells interact via a soft-sphere force potential with short-ranged attraction. Multiple contributions are calculated between every vertex and the closest point on the other cells edges. In addition, the cell cycle introduces growth of the bacteria until it divides in the middle into two new cells. This growth is downregulated by an increasing number of neighboring cells. Cells are placed inside the left-hand side of an elongated box with reflective boundary conditions. Their colors range from green for fast growth to blue for dormant cells.

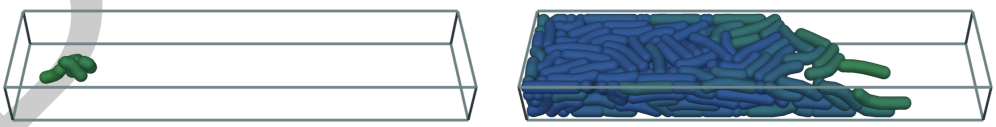

Figure 2: The bacteria extend from the initial placement in the left side towards the right side. Their elongated shape and the confined space favour the orientation facing along the growth direction.

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Branching of *Bacillus Subtilis*

Spatio-temporal patterns of bacterial growth such as in *Bacillus Subtilis* have been studied for numerous years (Kawasaki et al., 1997; Matsushita et al., 1998). Cells are modeled as soft spheres which take up nutrients from the domain. By consuming intracellular nutrients, the cell grows continuously and divides upon reaching a threshold. Cells are initially placed inside a centered square after which they grow outwards into the nutrient-rich area. They are colored bright purple while they are actively growing and dark when not subject to growth anymore. A lighter color in the outer domain indicates that more nutrients are available while a dark color signifies a lack thereof.

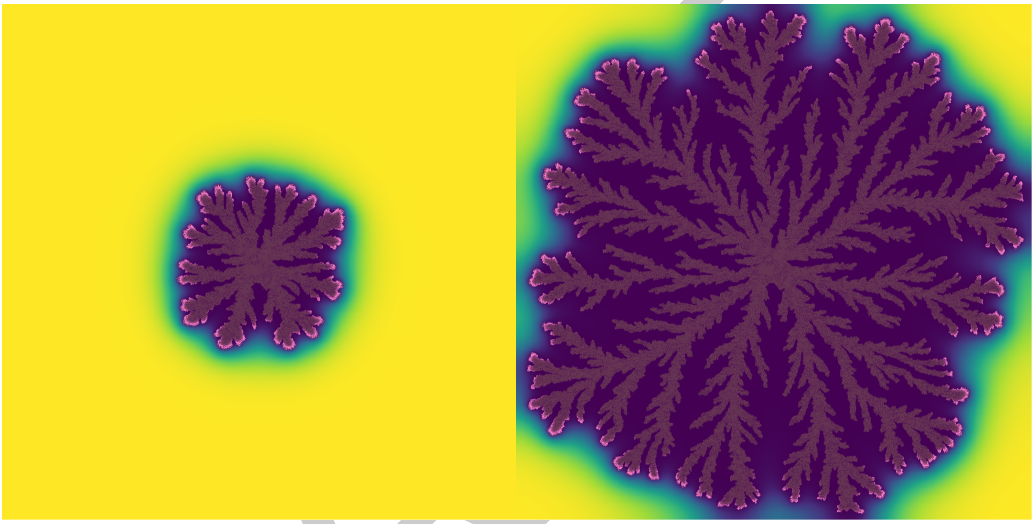

Figure 3: The bacterial colony grows outwards towards the nutrient-rich parts of the domain thus forming branches in the process.

Semi-Vertex Model for Epithelial and Plant Cells

Vertex models are actively being used to describe mechanical properties of plant cells (Merks et al., 2011) or organoid structures of epithelial cells (Barton et al., 2017; Fletcher et al., 2014). We represent cells by a polygonal collection of vertices connected by springs. An inside pressure pushes vertices outwards, creating perfect hexagonal cells. Cells are attracting each other but whenever two polygons overlap, a repulsive force acts. They are placed in a perfect hexagonal grid such that edges and vertices align and assigned growth rates from a uniform distribution.

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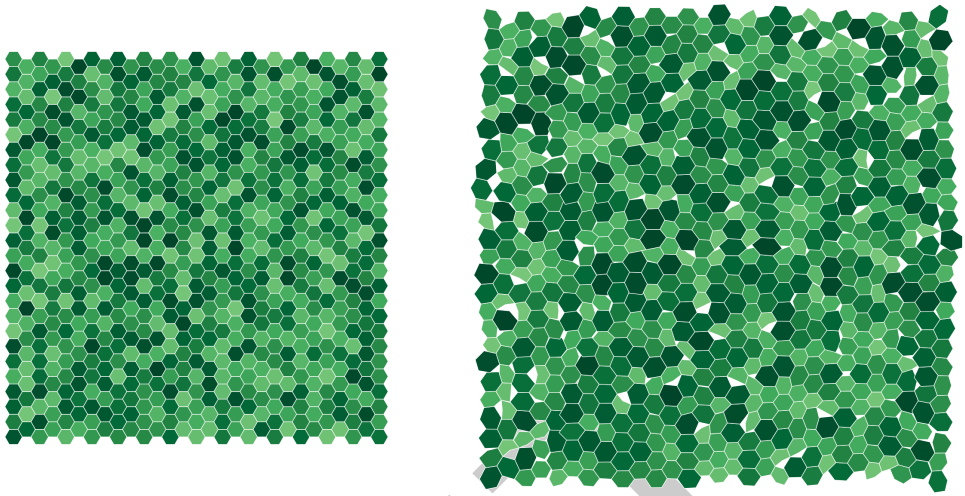

Figure 4: During growth the cells push on each other thus creating small spaces in between them as the collection expands. These forces also lead to deviations in the otherwise perfect hexagonal shape.

Further Information

The full documentation including guides, all examples from above and more is available at cellular-raza.com. `cellular_raza` can also be used as a numerical backend together with the `pyo3` and `maturin` konsti (2025) crates.

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References

Abar, S., Theodoropoulos, G. K., Lemarinier, P., & O'Hare, G. M. P. (2017). Agent based modelling and simulation tools: A review of the state-of-art software. *Computer Science Review*, *24*, 13–33. https://doi.org/10.1016/j.cosrev.2017.03.001

Barton, D. L., Henkes, S., Weijer, C. J., & Sknepnek, R. (2017). Active vertex model for cell-resolution description of epithelial tissue mechanics. *PLOS Computational Biology*, *13*(6), e1005569. https://doi.org/10.1371/journal.pcbi.1005569

Billaudeau, C., Chastanet, A., Yao, Z., Cornilleau, C., Mirouze, N., Fromion, V., & Carballido-López, R. (2017). Contrasting mechanisms of growth in two model rod-shaped bacteria. *Nature Communications*, *8*(1). https://doi.org/10.1038/ncomms15370

Breitwieser, L., Hesam, A., Montigny, J. de, Vavourakis, V., Iosif, A., Jennings, J., Kaiser, M., Manca, M., Di Meglio, A., Al-Ars, Z., Rademakers, F., Mutlu, O., & Bauer, R. (2021). BioDynaMo: a modular platform for high-performance agent-based simulation. *Bioinformatics*, *38*(2), 453–460. https://doi.org/10.1093/bioinformatics/btab649

Cess, C. G., & Finley, S. D. (2022). Multiscale modeling of tumor adaption and invasion following anti-angiogenic therapy. *Computational and Systems Oncology*, *2*(1). https://doi.org/10.1002/cso2.1032

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Cooper, F., Baker, R., Bernabeu, M., Bordas, R., Bowler, L., Bueno-Orovio, A., Byrne, H., Carapella, V., Cardone-Noott, L., Cooper, J., Dutta, S., Evans, B., Fletcher, A., Grogan, J., Guo, W., Harvey, D., Hendrix, M., Kay, D., Kursawe, J., … Gavaghan, D. (2020). Chaste: Cancer, heart and soft tissue environment. *Journal of Open Source Software*, *5*(47), 1848. https://doi.org/10.21105/joss.01848

Datseris, G., Vahdati, A. R., & DuBois, T. C. (2022). Agents.jl: A performant and feature-full agent-based modeling software of minimal code complexity. *SIMULATION*, 003754972110688. https://doi.org/10.1177/00375497211068820

Delile, J., Herrmann, M., Peyriéras, N., & Doursat, R. (2017a). A cell-based computational model of early embryogenesis coupling mechanical behaviour and gene regulation. *Nature Communications*, *8*(1). https://doi.org/10.1038/ncomms13929

Delile, J., Herrmann, M., Peyriéras, N., & Doursat, R. (2017b). A cell-based computational model of early embryogenesis coupling mechanical behaviour and gene regulation. In *Nature Communications* (No. 1; Vol. 8). Springer Science; Business Media LLC. https://doi.org/10.1038/ncomms13929

Fletcher, A. G., Osterfield, M., Baker, R. E., & Shvartsman, S. Y. (2014). Vertex models of epithelial morphogenesis. *Biophysical Journal*, *106*(11), 2291–2304. https://doi.org/10.1016/j.bpj.2013.11.4498

Graner, F., & Glazier, J. A. (1992). Simulation of biological cell sorting using a two-dimensional extended potts model. *Physical Review Letters*, *69*(13), 2013–2016. https://doi.org/10.1103/physrevlett.69.2013

Kang, S., Kahan, S., McDermott, J., Flann, N., & Shmulevich, I. (2014). Biocellion : Accelerating computer simulation of multicellular biological system models. *Bioinformatics*, *30*(21), 3101–3108. https://doi.org/10.1093/bioinformatics/btu498

Kawasaki, K., Mochizuki, A., Matsushita, M., Umeda, T., & Shigesada, N. (1997). *Modeling Spatio-Temporal Patterns Generated byBacillus subtilis*. https://doi.org/10.1006/jtbi.1997.0462

konsti. (2025). *maturin: Build and publish crates with pyo3, cffi and uniffi bindings as well as rust binaries as python packages* (Version 1.8.2). https://github.com/pyo3/maturin

Matsushita, M., Wakita, J., Itoh, H., Ràfols, I., Matsuyama, T., Sakaguchi, H., & Mimura, M. (1998). *Interface growth and pattern formation in bacterial colonies*. https://doi.org/10.1016/S0378-4371(97)00511-6

Merks, R. M. H., Guravage, M., Inzé, D., & Beemster, G. T. S. (2011). VirtualLeaf: An open-source framework for cell-based modeling of plant tissue growth and development. *Plant Physiology*, *155*(2), 656–666. https://doi.org/10.1104/pp.110.167619

Mogilner, A., & Manhart, A. (2016). Agent-based modeling: Case study in cleavage furrow models. *Molecular Biology of the Cell*, *27*(22), 3379–3384. https://doi.org/10.1091/mbc.e16-01-0013

Pleyer, J., & Fleck, C. (2023). Agent-based models in cellular systems. *Frontiers in Physics*, *10*. https://doi.org/10.3389/fphy.2022.968409

PyO3 Project and Contributors. (n.d.). *PyO3*.

Steinberg, M. S. (1963). Reconstruction of tissues by dissociated cells: Some morphogenetic tissue movements and the sorting out of embryonic cells may have a common explanation. *Science*, *141*(3579), 401–408. https://doi.org/10.1126/science.141.3579.401

Wilensky, U. (1999). *NetLogo* [Http://ccl.northwestern.edu/netlogo/]. Center for Connected Learning; Computer-Based Modeling. http://ccl.northwestern.edu/netlogo/

Young, K. D. (2006). The selective value of bacterial shape. *Microbiology and Molecular*

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Biology Reviews, *70*(3), 660–703. https://doi.org/10.1128/mmbr.00001-06

Zapun, A., Vernet, T., & Pinho, M. G. (2008). The different shapes of cocci. *FEMS Microbiology Reviews*, *32*(2), 345–360. https://doi.org/10.1111/j.1574-6976.2007.00098.x

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